Materials, stress & strain

ENGINEERING WITH MR KELLY

Higher · Mechanisms & structures · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ choose and justify a material from a data table
- ☐ calculate direct stress and strain
- ☐ calculate Young's modulus
- ☐ label a stress–strain graph: limit of proportionality, yield, UTS, break
- ☐ calculate factor of safety and strain energy
- ☐ explain how strain gauges are used to measure strain

I DO — Worked example

Read how a full-mark answer is built

A steel rod of diameter 10 mm carries a tensile load of 5 kN.

1 Calculate the stress in the rod. (3 marks)

Full-mark answer each ✓ is one mark

$A = \pi \times 5^2 = 78.5\ mm^2$ ✓. $\sigma = F \div A = 5000 \div 78.5$ ✓ $= 64\ N/mm^2$ ✓.

Why marks are not awarded

"$\sigma = 5 \div 78.5 = 0.064\ N/mm^2$"

2 marks: kN not converted to N.

WE DO — Guided practice

Use the starters and word bank to help you

A 2 m wire stretches by 1.2 mm.

2 Calculate the strain. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Strain has units of...
☐ N/mm² ☐ mm ☐ None ☐ N

b) A material that breaks with almost no plastic region is...
☐ Ductile ☐ Brittle ☐ Malleable ☐ Elastic

c) UTS 430 N/mm² with a factor of safety of 3 gives a working stress of about...
☐ 143 N/mm² ☐ 1290 N/mm² ☐ 433 N/mm² ☐ 3 N/mm²

d) Which justification scores?
☐ It is strong, so it is a good choice for the boat ☐ It is cheap and easy for the builder to get hold of ☐ It is a good material that will last a long time ☐ It is corrosion resistant, so it won't degrade in sea water

4 Key terms: write the letter of the matching description next to each term.

1. **Fatigue**	____	**A** ε = Δl ÷ l (no units)
2. **Stress**	____	**B** UTS ÷ working stress
3. **Strain**	____	**C** σ = F ÷ A (N/mm²)
4. **Justifying a material**	____	**D** Failure from repeated loading, below the UTS.
5. **Young's modulus**	____	**E** Name the property and link it to the job.
6. **Factor of safety**	____	**F** E = stress ÷ strain

A hydrofoil mast must be strong, light and resist corrosion in seawater.

5 Choose a suitable material from aluminium, mild steel and carbon fibre, and justify your choice. (2 marks)

A tie has a working stress of 85 N/mm². Its material has a UTS of 430 N/mm².

6 Calculate the factor of safety. (2 marks)

A structural engineer recommends a high factor of safety for a footbridge.

7 Give two reasons for the advice. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?